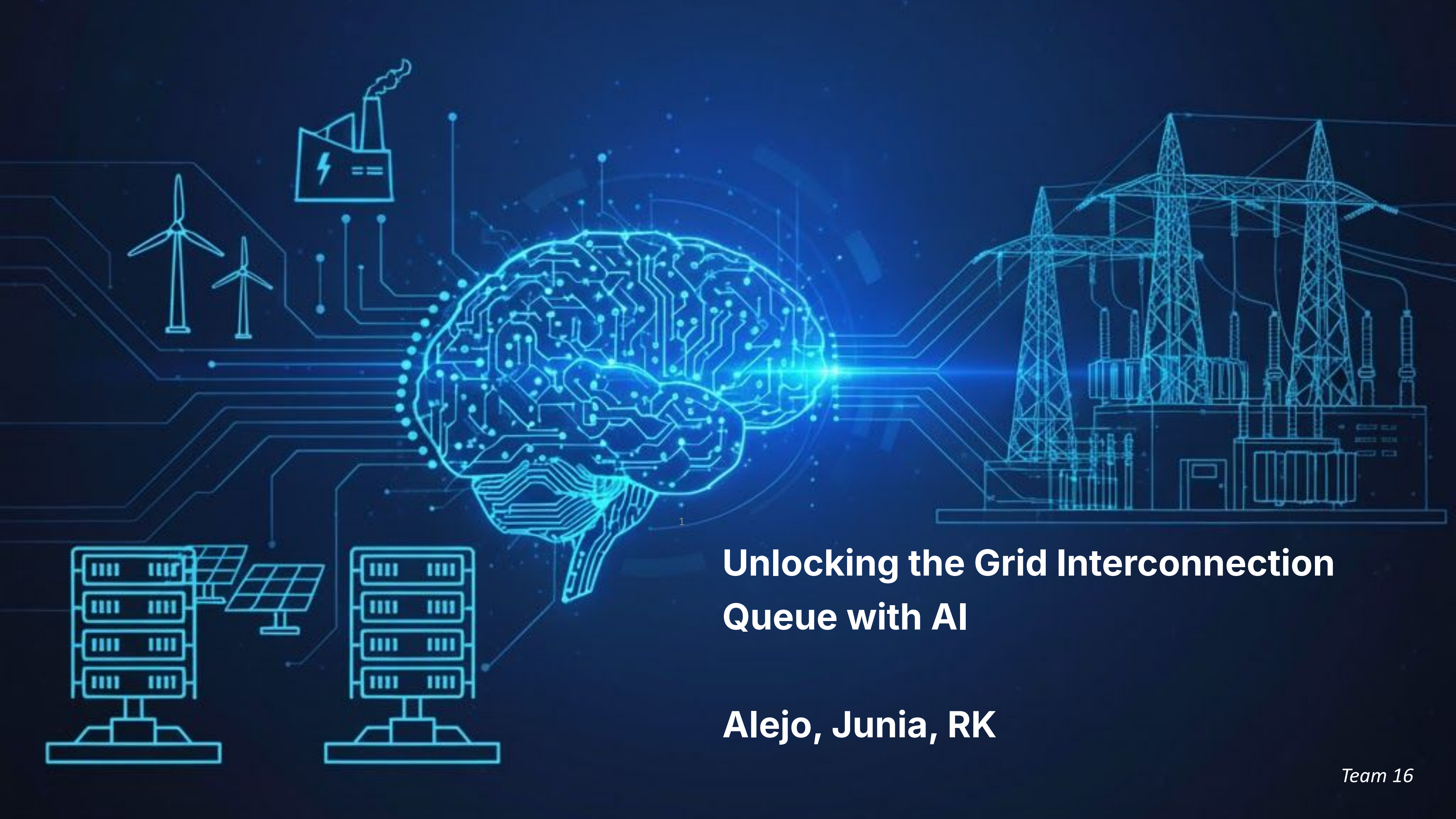




Unlocking the Grid Interconnection Queue with AI

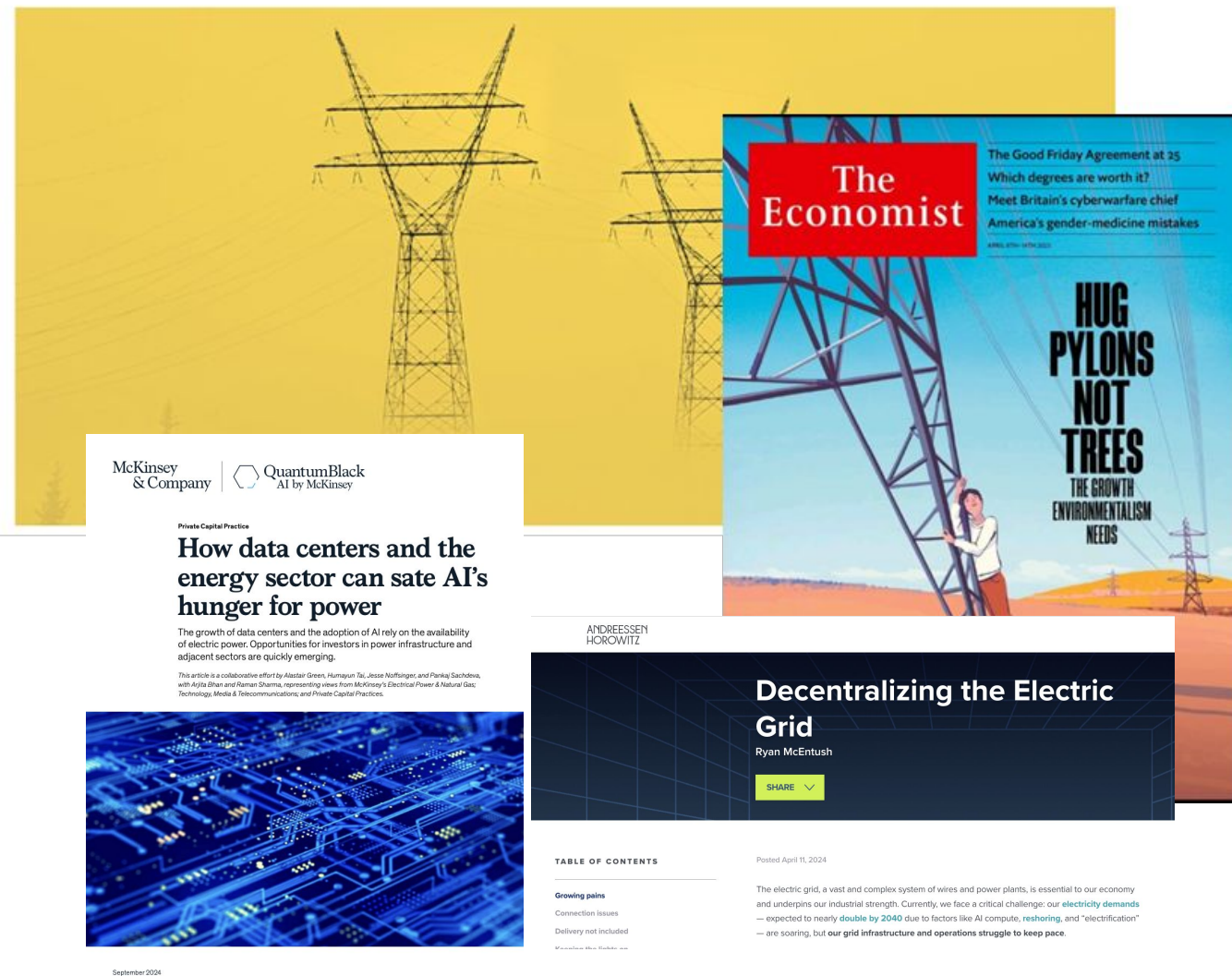
Alejo, Junia, RK

Boost of Interconnection Requests (IR) to the Power Grid

ON THE GRID

The surprising key to a clean energy future

If you care about climate change, you should care about transmission.

By **Bill Gates** | January 24, 2023 • 5 minute read

Expectation:



Interconnection Queue Costs the Global Economy \$50-80B per Year

Reality:



80% of projects withdraw
during 3-5 year queue process

\$2-3B* addressable business potential

3 The annual global revenue opportunity for providers of interconnection-specific services and tools (e.g., feasibility studies, system impact analyses, compliance modeling, queue management platforms)

U.S. \$800-900M*

AI-solvable challenges in the grid interconnection process

AI enables targeted solutions across four critical domains, each with defined implementation timelines and proven methodologies.



Document Processing
Challenge: Application review & validation



Technical Studies
Challenge: Grid impact analysis & modeling

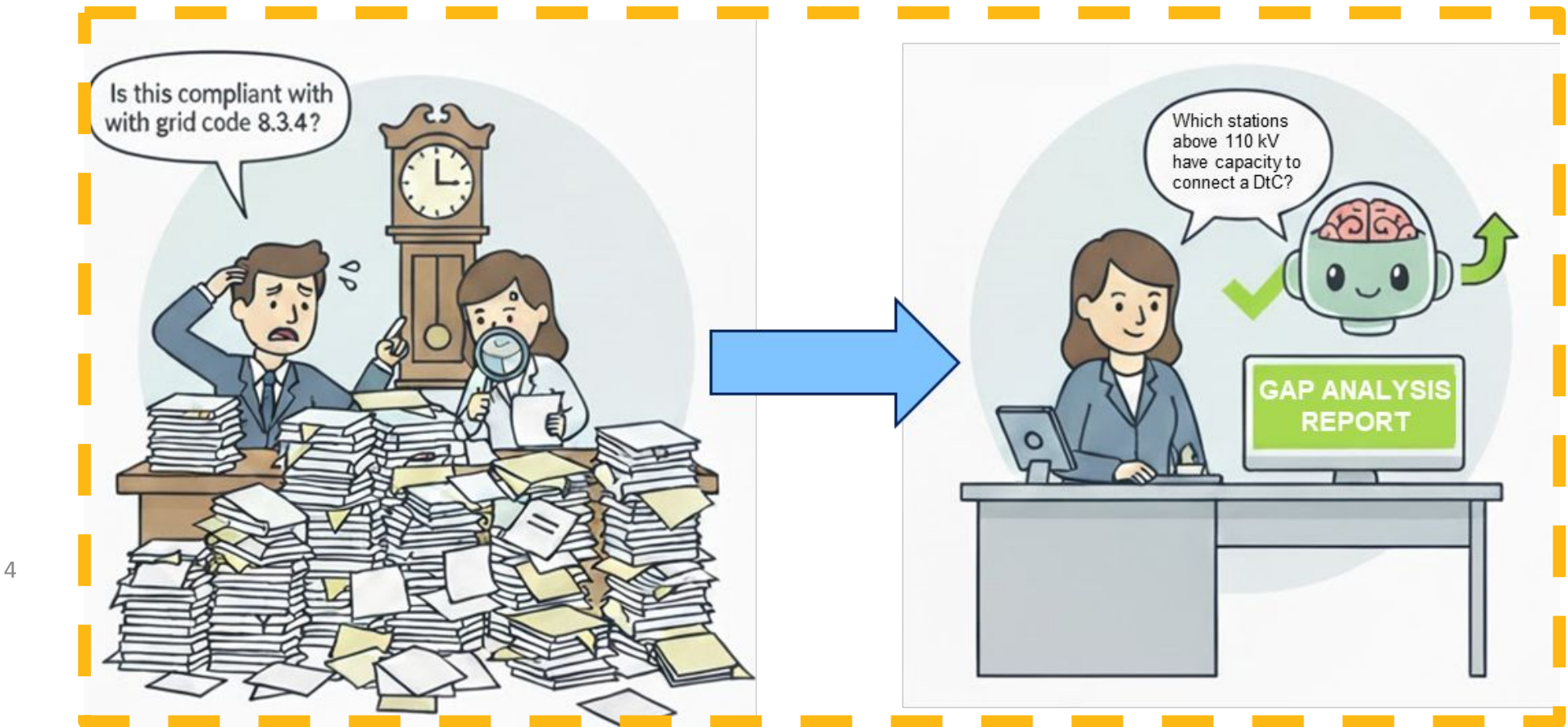


Grid Expansion Planning
Challenge: Optimal grid expansion



Decision Support
Challenge: Coordination between stakeholders

CAPSTONE SCOPE

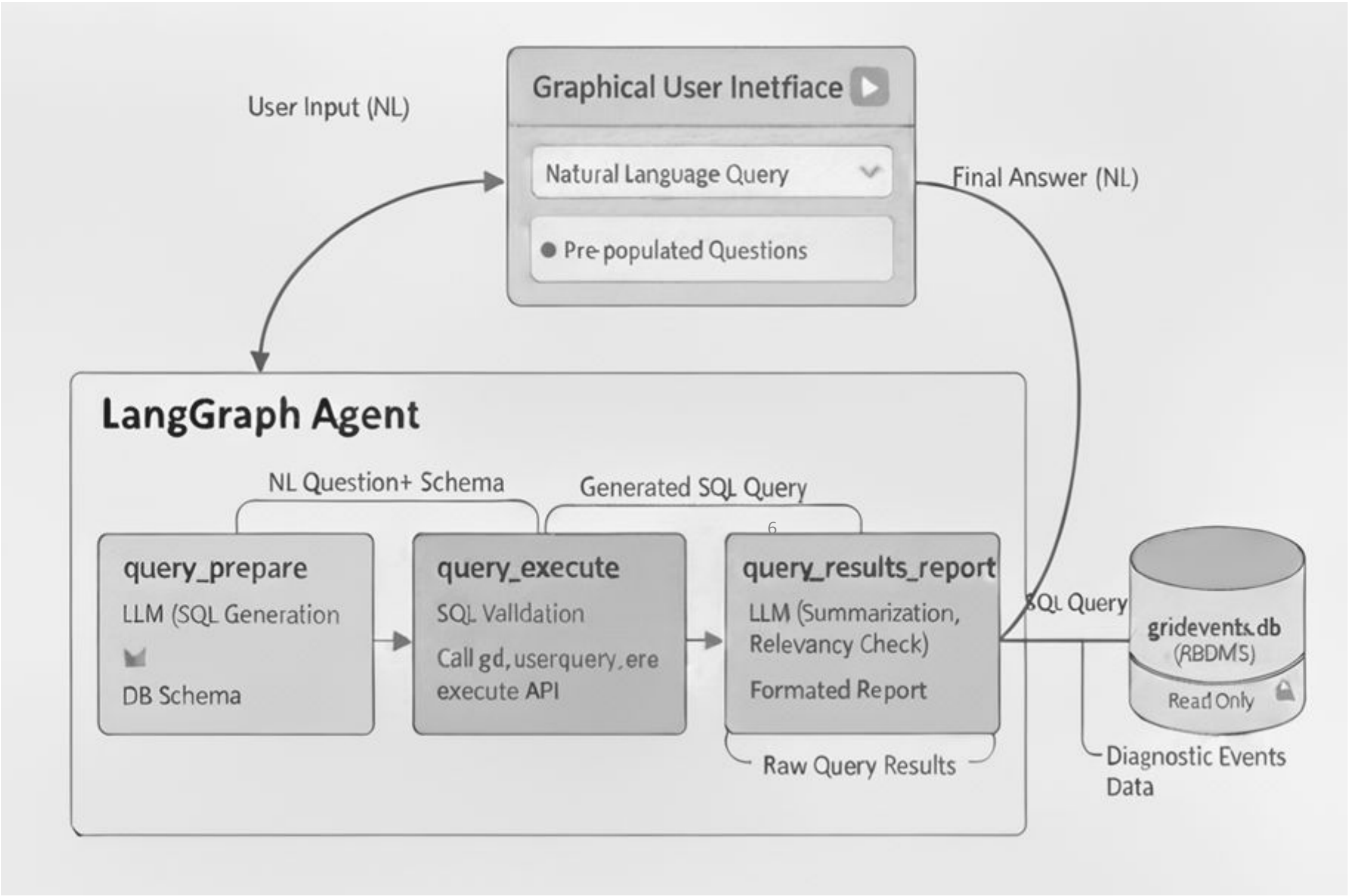


Power Grids Interconnection Queue: Compliance AI Agent

Problem-First based Capstone Iterations:

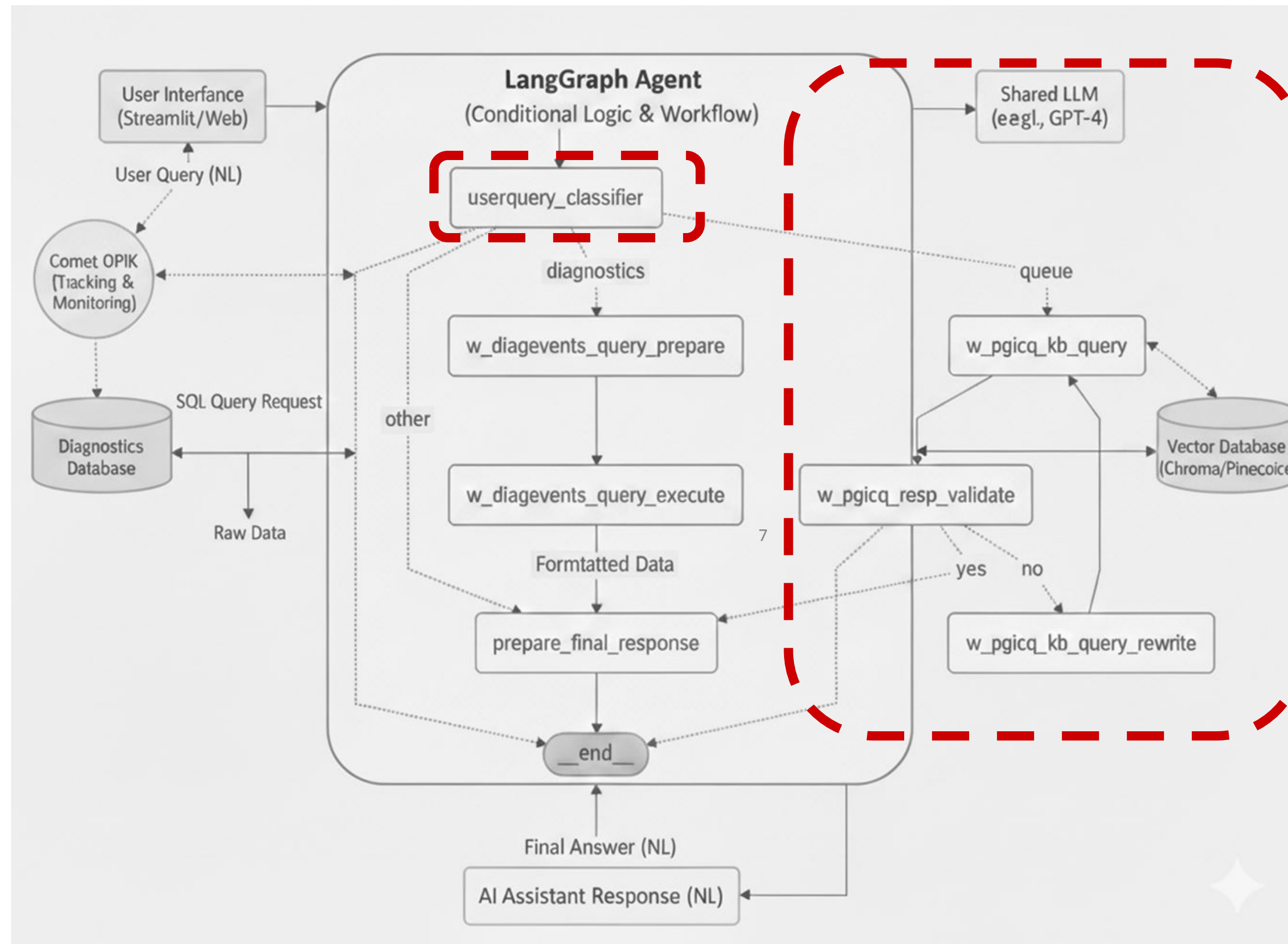
- Iteration 1
Grid diagnostics
- Iteration 2
Iteration 1 + Grid interconnection process analyzer
- Iteration 3
Iteration 2 + Gap analysis of interconnection requests

Compliance AI agent Iteration 1: Grid Diagnostics



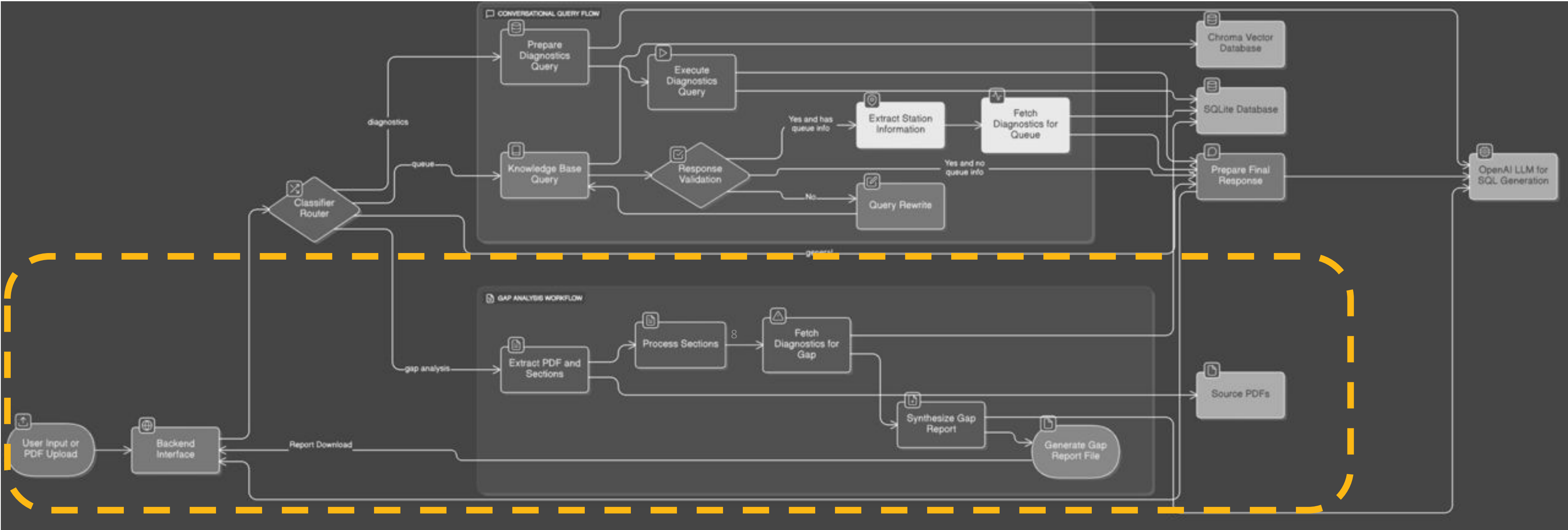
- OpenAI gpt-5-mini
- Streamlit
- LangGraph
- SQLite3
- Comet Opik

Compliance AI agent Iteration 2: Grid Interconnection Process Analyzer



- OpenAI gpt-5-mini
- Streamlit
- LangGraph
- SQLite3
- Comet Opik
- Chroma DB
- OpenAI Embeddings

Compliance AI agent Iteration 3: Gap Analysis of Interconnection Requests



Tech stack as Iteration 2

Evals & Measures for Compliance AI agent: Iteration 1

Iteration #1: Evals and measurements						
Component/ Module	Measurement	Metric	Actual / Measured Trial - 1	Actual / Measured Trial - 2	Actual / Measured Trial - 3	Avg of trial 1, 2 & 3
SQL query generation	Syntax Correctness	Execution Success Rate (% queries that run without errors)	1	1	1	1
SQL query generation	Query Accuracy	Precision/Recall on retrieved data	1	1	1	1
SQL query generation	Query Efficiency	Query execution time	0.005	0.004	0.007	0.0053333
SQL query generation	Edge Cases	False positive rate (returns wrong data)	0	0.5	1	0.5
LLM Prompt Effectiveness	Prompt Reliability	Consistency score (same question → same SQL structure)	1	1	1	1
LLM Prompt Effectiveness	Instruction Following	Schema adherence rate	1	1	0.5	0.8333333
LLM Prompt Effectiveness	Output Format	Format compliance (no markdown, no extra text)	1	1	1	1
LLM Prompt Effectiveness	Hallucination Rate	Hallucination detection (non-existent columns/tables)	1	1	1	1
Response Synthesis Quality	Accuracy	Factual accuracy (all numbers match DB)	1	1	1	1
Response Synthesis Quality	Completeness	Answer completeness score	1	1	1	1
Response Synthesis Quality	Clarity	Readability score	1	1	1	1
Response Synthesis Quality	Human Eval rubric	Relevance (1 to 5)	5	5	5	5
Response Synthesis Quality	Human Eval rubric	Accuracy (1 to 5)	5	5	1	3.6666667
Response Synthesis Quality	Human Eval rubric	Clarity (1 to 5)	5	5	5	5
End-to-End Pipeline Performance	Latency: Total time from question to answer	P50, P95, P99	9.04	9.35	7.88	8.7566667
End-to-End Pipeline Performance	Success Rate: % of queries that complete successfully	Overall success rate	1	1	1	1
LLM-Specific Metrics (gpt-5-mini)	Token Usage	Tokens per query (input + output)	691	972	493	718.66667
LLM-Specific Metrics	Token Usage	Cost per query	0.001	0.001	0.001	0.001
LLM-Specific Metrics	Response Quality	Accuracy by model	1	1	1	1
LLM-Specific Metrics	Response time	Response time in sec	9	9.3	6.1	8.1333333

Evals & Measures for Compliance AI agent: Iteration 2 & 3

Iteration #2: Evals and measurements						
Component/ Module	Category	Metric	Actual / Measured Trial - 1	Actual / Measured Trial - 2	Actual / Measured Trial - 3	Avg of trial 1, 2 & 3
LLM Prompt Effectiveness on userquery classifier	Prompt Reliability	Consistency score	1	1	1	1
LLM Prompt Effectiveness on userquery classifier	Instruction Following	adherence rate to instructions	1	1	1	1
LLM Prompt Effectiveness on userquery classifier	Output Format	Format compliance (no markdown, no extra text)	1	1	1	1
LLM Prompt Effectiveness on userquery classifier	Hallucination Rate	Hallucination detection (non-existent data or tables)	1	1	1	1
RAG	better coverage	Maximum marginal relevance (balance relevance and diversity fetch_k=20): more docs	Performance gain BEFORE: - Query retrieval: 400-500ms (similarity search) - Validation: 2-3s (LLM call) - Synthesis: 3-5s (verbose prompt, 5000+ chars) TOTAL: 6-9 seconds per query AFTER: - Query retrieval: 400-500ms (MMR - same speed) - Validation: SKIPPED (0s for successful queries) - Synthesis: 2-3s (concise prompt, 3000 chars max) TOTAL: 2.5-4 seconds per query SPEEDUP: 2-3x faster (60-70% time reduction) Cost Savings: BEFORE (per query): - Retrieval: \$0.0001 (embedding) - Validation: \$0.003 (LLM call) - Synthesis: \$0.005 (verbose prompt, 5000+ chars) TOTAL: ~\$0.008 per query AFTER (per query): - Retrieval: \$0.0001 (embedding) - Validation: \$0 (skipped) - Synthesis: \$0.003 (concise prompt, 3000 chars) TOTAL: ~\$0.003 per query SAVINGS: 60% cost reduction Quality Improvements: Better diversity (MMR) Top docs get more space (smart truncation) Less redundant information More focused answers Better source coverage			
RAG	reuse existing data	skip the validation every time if first retrieve successful.: saves 2-3 seconds				
RAG	Optimization on validation data	sent only first 200 characters for validation instead of full chunk. Reduced tokens, time and cost.				
RAG	relevance check optimization	reduced relevance check prompt to use less tokens and faster response				
RAG	better quality	relevance check instead of having same size for all searchable docs, top docs get more chunk (500), bottom docs get less size chunk (200). Saved tokens and time too				
RAG	Optimized the cotext usage	Instead of sending more content to LLM like up to 10K reduced to 3k size.				
RAG	relevance check optimization	reduced relevance check prompt to use less tokens and faster response				
			1	1	1	1
Response Synthesis Quality	Accuracy	Factual accuracy (all numbers match DB)	1	1	1	1
Response Synthesis Quality	Completeness	Answer completeness score	1	1	1	1
Response Synthesis Quality	Clarity	Readability score	1	1	1	1
Response Synthesis Quality	Human Eval rubric	Relevance (1 to 5)	5	5	5	5
Response Synthesis Quality	Human Eval rubric	Accuracy (1 to 5)	5	5	5	5
Response Synthesis Quality	Human Eval rubric	Clarity (1 to 5)	5	5	5	5
End-to-End Pipeline Performance	Latency: Total time from question to answer	P50, P95, P99	19.84	25.2	17.8	20.94666667
End-to-End Pipeline Performance	Success Rate: % of queries that complete successfully	Overall success rate	1	1	1	1
LLM-Specific Metrics (gpt-5-mini)	Token Usage	Tokens per query (input + output)	2587	2654	2338	2526.333333
LLM-Specific Metrics	Token Usage	Cost per query	0.002	0.001	0.002	0.001666667
LLM-Specific Metrics	Response Quality	Accuracy by model	1	1	1	1
LLM-Specific Metrics	Response time	Response time in sec	19	23	17	19.66666667

Performance gain

BEFORE:

- Query retrieval: 400-500ms (similarity search)
 - Validation: 2-3s (LLM call)
 - Synthesis: 3-5s (verbose prompt, 5000+ chars)
- TOTAL: 6-9 seconds per query

AFTER:

- Query retrieval: 400-500ms (MMR - same speed)
 - Validation: SKIPPED (0s for successful queries)
 - Synthesis: 2-3s (concise prompt, 3000 chars max)
- TOTAL: 2.5-4 seconds per query

SPEEDUP: 2-3x faster (60-70% time reduction)

Cost Savings:

BEFORE (per query):

- Retrieval: \$0.0001 (embedding)
 - Validation: \$0.003 (LLM call)
 - Synthesis: \$0.005 (verbose prompt, 5000+ chars)
- TOTAL: ~\$0.008 per query

AFTER (per query):

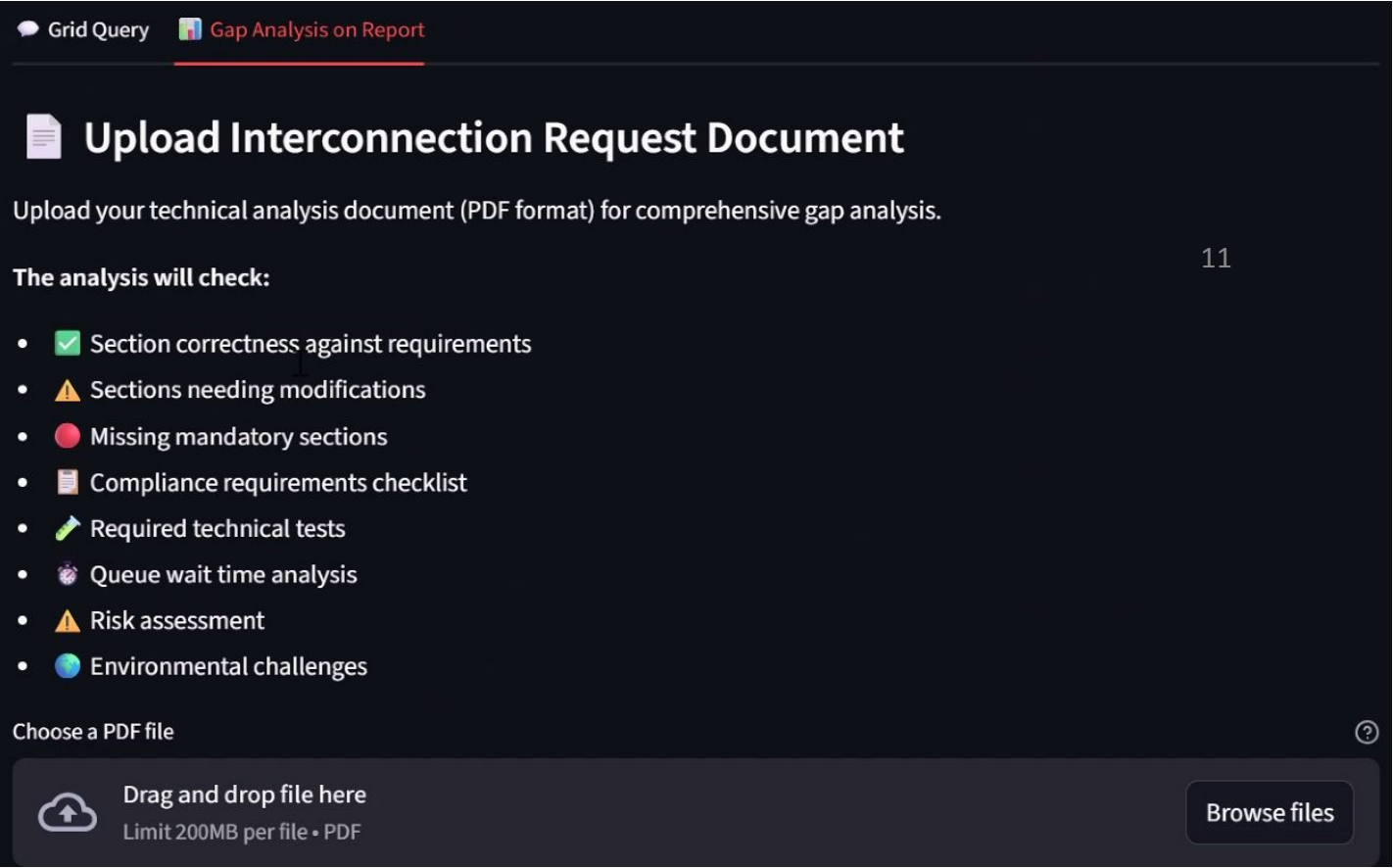
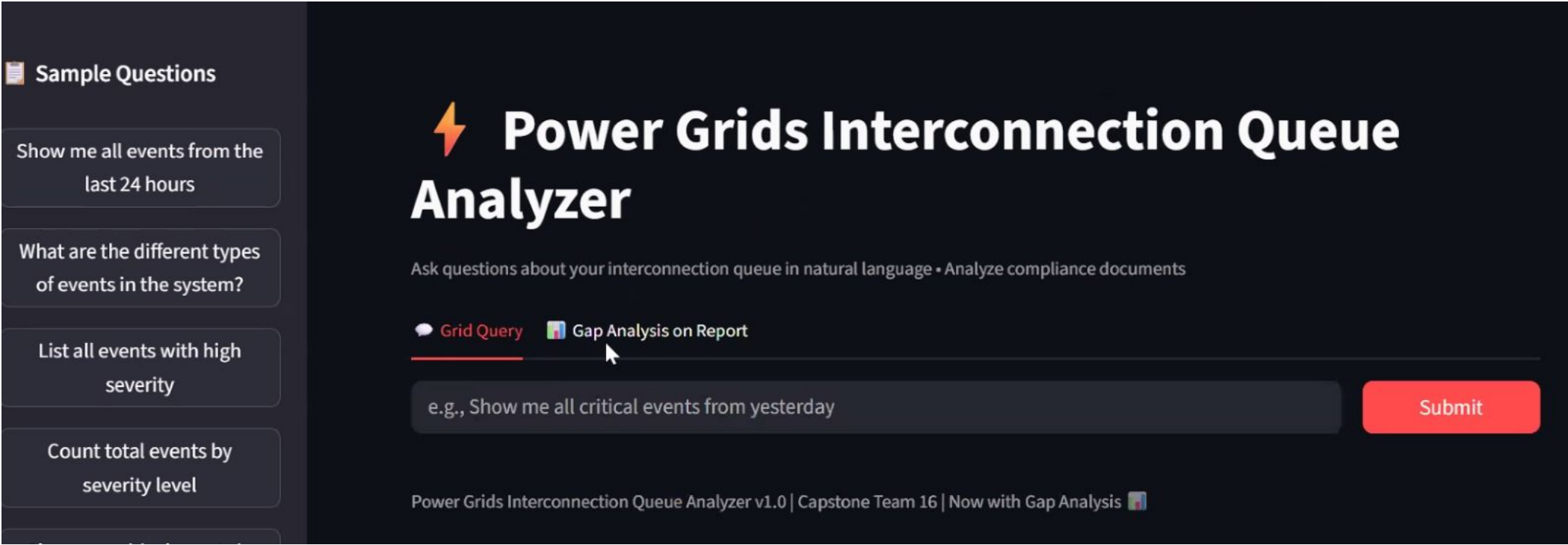
- Retrieval: \$0.0001 (embedding)
 - Validation: \$0 (skipped)
 - Synthesis: \$0.003 (concise prompt, 3000 chars)
- TOTAL: ~\$0.003 per query

SAVINGS: 60% cost reduction

Quality Improvements:

- Better diversity (MMR)
- Top docs get more space (smart truncation)
- Less redundant information
- More focused answers
- Better source coverage

Demo



GAP ANALYSIS REPORT Interconnection Request Technical Analysis	
Station:	Palo Alto 110 kV Substation
Region:	Palo Alto / California
Country:	United States
Request Type:	New connection (interconnection request for new 80 MW load)
Report Generated:	2025-11-06 19:36:06
Analysis ID:	20251106_193606

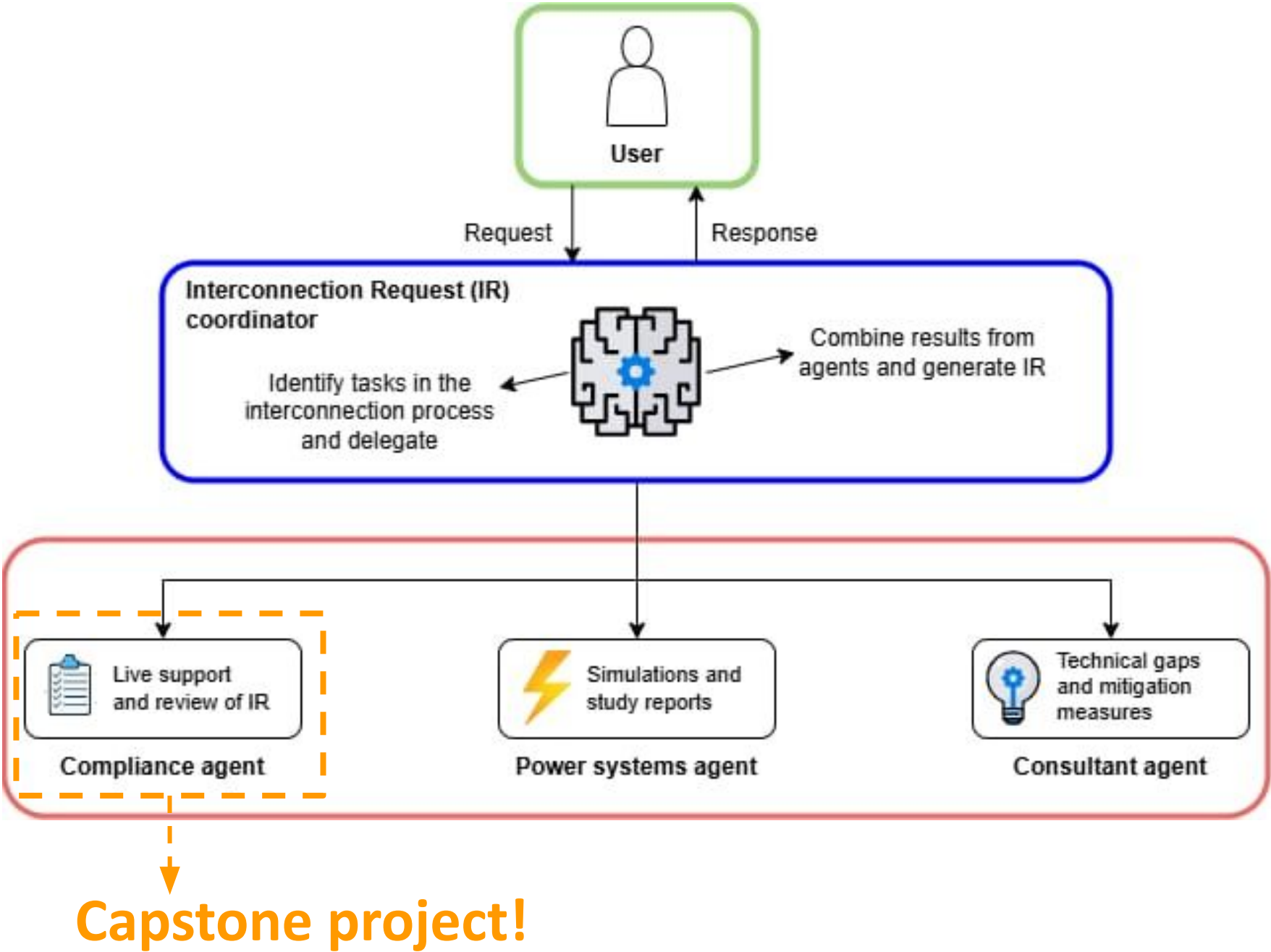
This report was generated by the Power Grids Interconnection Queue Analyzer
Capstone Team 16 - Grid Chat System

Our Idea: An AI Turnkey Solution to Run the Interconnection Process End to End

Goal:



12



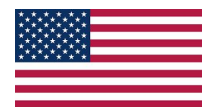
Thank You for Your Attention!



RK
K.V. Ramakrishna Rao



Junia¹³
Junia Compostella



Alejo
Alejandro Duque



Any questions?